BHARATH H V

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Professional Summary

Mechanical Design Engineer specializing in Aircraft Structures with hands-on experience in design and drafting of 4.5 Gen fighter aircraft LCA MkII and Airbus A320 concession projects. Skilled in structural defect resolution, engineering documentation, and end-to-end T0–T100 analysis.

Technical Skills

Design & CAD: CATIA V5 (Part Design, Drafting, GSD MBD, FTA) DMU
Standards & Methods: GD&T (ASME Y14.5 – fundamentals), SRM-based repair assessment
Engineering Tools: ENOVIA, SAP (PAC, PE1)
Documentation: Concessions, ESDM, ENCA, repair sketches and drawings

Professional Experience

AXISCADES Engineering Technologies Pvt. Ltd. Bengaluru
Assistant Engineer - Aircraft Structures. Dec 2023 – Present

- Performed end-to-end analysis of structural non-conformities in fuselage primary and secondary structure components.
- Assessed damage acceptability using approved design data and Structural Repair Manual (SRM) guidelines.
- Designed repair solutions and minor structural parts such as shims, bushes, brackets, and reinforcements.
- Prepared engineering documents including Concessions, ESDM, and ENCA with full technical justification.
- Supported aircraft assembly by resolving production issues while ensuring structural integrity and compliance.
- Acted as T100 design signatory for assigned repair and concession tasks.

HAL – Aircraft Research and Design Centre (ARDC) Bengaluru
Design Engineer – Structures Jun 2023 – Nov 2023

- Collaborated with multifunctional teams to design and draft structural aircraft components for a forthcoming 4.5 Gen fighter aircraft.
- Developed detailed sub-assembly drawings for frames, bulkheads, and longerons while adhering to Design for Manufacture and Assembly (DFMA) principles.
- Provided engineering support to production by addressing structural defects and assembly concerns.
- Selected materials and surface finishes based on functional requirements, load conditions, availability, and weight optimization.
- Implemented Model-Based Definition (MBD) using GD&T standards to ensure dimensional accuracy and manufacturability for parts.
- Selected suitable fasteners based on load paths, design specifications, and mating material characteristics.

Education

Bachelor of Engineering – Mechanical Engineering
Malnad College of Engineering, Hassan
(Autonomous-VTU) 2019 – 2022

Certifications

- CATIA V5 Basics – Great Learning
- Welding Application Technology – NPTEL (IIT Guwahati)